Analysis of Saccadic Eye Movements and Reaction Times

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A. Objective of the project

The objective of this study is to determine human reaction times by comparing the results obtained through the Pupil Core system with those measured using the High-Speed camera, which serves as the reference system. The aim is to quantify the deviation between the two methods and assess the reliability of the Pupil Core algorithm. *This study highlights the difficulties in achieving optimal reaction time measurements.*

B. Methodology

Materials

The study utilized:

- **High-Speed eye-tracking system (***High-Speed camera e Pupil Core)***:** To capture saccadic movements and pupil positions.
- **MATLAB:** For data processing, signal analysis, and implementation of Monte Carlo simulations.
- **Statistical libraries:** To simulate uncertainty distributions and analyze variability in the results.

Methods

1. **Data Acquisition:** Eye movement data were recorded during controlled visual stimuli presentations.
2. **Signal Processing:** Position and timestamp profiles were analyzed to extract saccadic onset and offset times.
3. **Monte Carlo Simulation:** Distributions of input parameters, such as detection thresholds and noise, were modeled to estimate reaction time variability.

Validation

Validation was carried out by comparing simulation outputs with recorded empirical data. Reference datasets from previous studies on saccadic response times were used to benchmark the model's accuracy and ensure reliable results. One of these reference data points from other studies included the world record reaction time of 120 ms. Therefore, all measurements equal to or below 120 ms were excluded.

Experimental Setup

Experiments were conducted in a laboratory setting with controlled lighting conditions. The eye-tracking device was calibrated for each participant to ensure accurate measurements of pupil position and movement. Data was collected using the software specific to each camera, including *Pupil Core*, *Pupil Player*, and the High-Speed camera program AIS (Promon U750). These datasets were subsequently processed and analyzed in MATLAB.

Experimental protocol

1. Participants were seated at a fixed distance from the stimulus display, after the calibration activity was implemented.
2. Visual stimuli consisted of a fixed cross always displayed at the center of the screen and a dot alternately appearing on the right and left side of the screen. The positions of the stimuli remained constant throughout the experiment.
3. Eye-tracking data were collected continuously during the session.

4. Reaction times were computed as the delay between stimulus onset and saccadic movement initiation.

C. Results

The analysis revealed that:

An uncertainty of *30,16 ms* was achieved for the project.

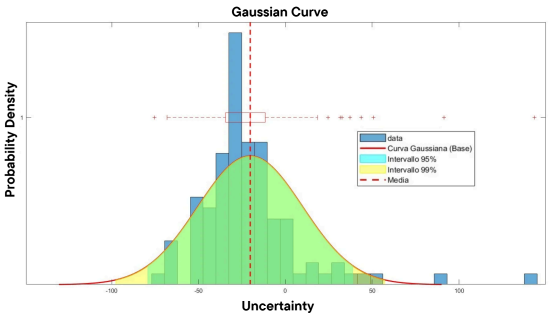


Figure 1 - The Gaussian Curve of uncertainty (Pupil compared with the High-Speed Camera)

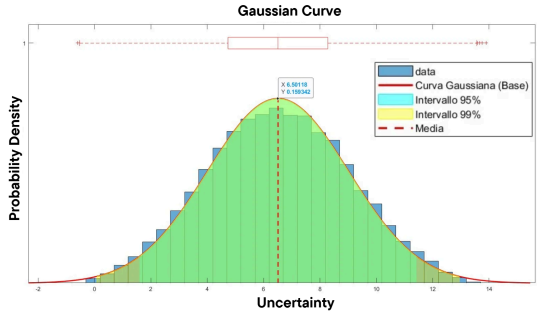


Figure 2 -The Gaussian uncertainty curve based on 30,000 reaction time simulations using Monte Carlo

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Description of the results

Monte Carlo simulations confirmed that algorithmic uncertainty slightly impacts the accuracy of the pupil system algorithm. This simulation specifically focused on estimating uncertainties arising from variations in detection thresholds and noise. However, the uncertainty of the overall program (High-Speed camera + Pupil Core) was not analyzed through Monte Carlo simulations.

The measurements obtained from both systems (High-Speed Camera & Pupil) cannot be considered fully compatible, as the differences observed in the intervals highlighted in the table (Table 1) and in the graph (Figure 3) are significant.

	H-S C	Pupil
Mean	0 ms	-20.28 ms
Contributory elements in uncertainty	$\sigma_{NTP} = 5.77\ ms$ $\sigma_{Promon} = 1.16\ ms$ $\sigma_{Manuale} = 4.24\ ms$	$\sigma_{NTP} = 5.77\ ms$ $\sigma_{PUPIL-IR} = 1.44\ ms$ $\sigma_{PUPIL-Ext} = 9.53\ ms$ $\sigma_{Algoritmo} = 2.5\ ms$ $\sigma_{Latenza} = 5\ ms$
Uncertainty	7.25 ms	30.16 ms
Interval (P=95%)	± 14.21 ms	[-80.50, 40.14] ms

Table 1 - Results

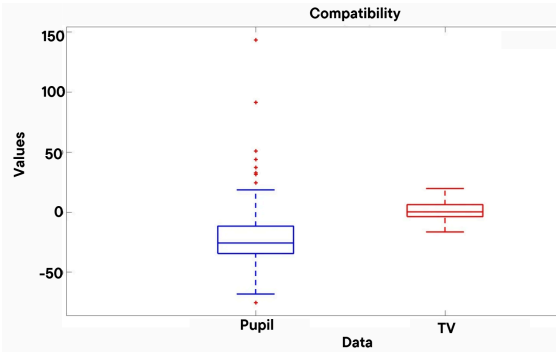


Figure 3 -Representation of the data collected using two boxplots: Pupil system (blue) and High-Speed Camera (red)

D. Extra:

The results showed an average reaction time of 278 ms, which significantly deviates from the optimal values reported in previous research (typically around 220 ms with less variability). This greater variability suggests that the experimental setup or external factors may have influenced the results.